Bounded execution of twist routes 1–3

Runs: 2026-10-21–22. PR: #2380.

The bounded panels produced executable trace, cover, divisor-class, equivalent-system, isogeny and rational-lift checks. The combined native IC panel recorded **1,920 executions: 1,704 verified logarithms and 216 empty-base errors**. All 1,704 verified records reproduced exactly in the native audit. **A calibrated IC speedup, a novel algorithm and a NIST-size benefit are not established.**

The original eighteen-experiment program remains broader than these executions. Three-gluing construction, complete higher-degree divisor factorization, broader solver grids, full accounting/calibration and independent-host replay remain open. These are explicit partial results, not a completed full campaign.

Inputs and execution

The frozen manifest contains two structural models, eight public-seeded A=-3 ordinary prime-order ladder models at field sizes 8, 10, 12 and 14 bits, and three even/composite controls. It records exact orders, generators, quadratic/cubic/sextic fields, tower conversions and rejected ladder candidates. Its SHA-256 is `8d490a36d65ebf906b93e04eeeb21ac5802f2a9382ef03271a4648fd4b78a4ed`. The canonical Rust identity constructor independently checked all 67 manifest models, including existing entries, against the registry and alias map. The registry gained 65 models; 34 of them are extension models.

All experiments used native Rust or native PARI/GP, Singular and msolve on the same Debian 13 x86_64 managed host. The host has a four-CPU quota and 16 GiB memory limit. Individual CAS limits and source/input freezes are in the versioned protocols and execution receipts. Runs are L0. No wall-clock comparison is admitted. NIST P-192/P-224/P-256/P-384/P-521 appear only in public-parameter applicability checks; no NIST DLP was executed.

Recorded results

- **Trace and rational-only controls:** nine exhaustive quadratic-extension censuses and 30 exact quartic/extension witness-set comparisons passed. For `icv1-fp6-tm7-fc20da94`, the 3,551 extension points give 67 trace images, with 53 lifts per trace. Repeated lifts do not become independent rows. There are 1,179 configured relation-census rows: 678 verified and 501 unsupported empty/undersized bases; identical bases are explicitly reused.
- **Covers and transfer:** exhaustive search of 27,140 translate parameters on ten fixtures found 11 actual twist complements, with at least one per fixture. Matching j or opposite trace alone was rejected. All ten selected covers passed executable pullback/norm tests, including all 3,551 rational classes on the smallest Jacobian and 256 sampled classes on each larger fixture: 5,855 classes total, plus 1,280 additivity/associativity samples. PARI independently reproduced all ten characteristic polynomials.
- **Three-gluing module filter:** a separately frozen PARI calculation on the public orders reproduces all ten toy E11 counts. Of the five NIST P-curves, only P-521 passes, with four determinant-minus-one anti-intertwiners over F3. P-192, P-224, P-256 and P-384 have zero. This necessary companion-module test supplies a construction candidate; the actual torsion/pairing bridge, polarized quotient, genus-two curve and divisor maps remain open. See the raw result and report.
- **Low-degree isogenies:** of 30 degree/fixture cells, three degree-seven maps were constructed and 27 cells failed the stated applicability filters. The first run supplied two algebraic duals. A separately frozen amendment also samples independent torsion with a repeated eigenvalue and requires a nonzero forward image. The new run supplies all three explicit duals; composition [7] is checked on every rational source point, with 24 scalar return controls. The finite lookup inverses remain separate toy controls.
- **Rational lifts:** 254 standard-model-height searches found points in 25 cases and three-term relations in five. The separate equal-original-height control found points in 76 cases and triples in seven. Square-parameter reductions and every projected relation passed the identical-base ordinary group census. No projected row lies outside that census. One rank interval remains unresolved; these are not certified full Mordell-Weil bases.

- **Conditional pairing control:** a separate registered supersingular model with r=11 and embedding degree two passed ten native Miller/bilinearity and independent PARI/source-log checks. This is a deliberately known-scalar positive control, with no ordinary/NIST transfer or speed claim.

These equivalence checks validate the implementations and remove duplicate or incorrect projection explanations. They do not measure whether an auxiliary family could find the same useful rational relations more cheaply.

Solver attempts and limits

Each attempt contains the same 1,930 configured cells: ten fixtures, eight public target seeds, arities 2–4, product/chain/Boolean-index/first-pair-symmetric encodings, grevlex/lex and ten A/A cells. The 240 distinct projected-set tests pass in each attempt; repetitions are not additional independent instances.

Attempt	Native complete in joined records	Singular dimensions	msolve q0 bases	Interpretation
V1	558	0 accepted	0	External input invalid: Unicode operators; 458 cell execution/parsing errors retained
V2	731	970	0	ASCII fixed; 768 MiB address cap crashes msolve on a tiny valid input
V3	730	968	1,096	Same cells at 4 GiB per engine; bounded native/q0/ideal checks only

V3 verifies 866 native and 940 msolve-q0 ideal equalities through independent Singular reductions. An ideal equality does not certify that a truncated basis is a complete Gröbner basis. Timeouts, missing outputs and unresolved calls remain incomplete. The one-second hard child limit and native 500 ms cooperative limit prevent interpreting completion-count differences as speed ratios across attempts.

The installed msolve signature mode rejects these nonhomogeneous inputs; a retained diagnostic also reports its different working characteristic. It supplies no admissible same-field signature baseline. Magma is unavailable. Kernel multiplication counts do not price complete solver cost. B=12/16 and the other projected-base families remain outside this initial solver panel.

Combined native IC panel

Spec, protocol, raw session, run report.

There are eight public one-target workloads per curve, 24 arms, one L0 round and zero warmup. The panel includes all eight trace/cover/symmetry combinations, four identical-base blocked-MITM controls, nine ordinary IC configurations (three sizes by three oracles), matched negation rho, frozen rho and an A/A rho control. Every supported cell returned the correct logarithm. The 216 errors are the empty trace/cover intersection on nine curves, repeated across the two solver arms and their same-base control. They remain in the attempt ledger and are not zero-cost successful solves.

The native audit passed for all 1,920 records, reproducing all 1,704 verified records with zero problems. Its SHA-256 is `64da13d34c144d2a6d05d13d1a1208ca75684683608eecaa0f0d76fc9b3961cb`. This is same-host replay.

The adapters rebuild the base and discover the cover for each cold target. Signed lifting and verification use the counted curve law. Construction, reference-arithmetic calls, F4 multiplications, symbolic terms

and root extraction remain explicit unpriced units. Reference-arithmetic calls are high-level units, not complete primitive-field or instruction counts.

The boundary/pricing table gives every curve/arm's S, generic negation floor, matched rho ratios and equal 0/0.1/1 GAE sensitivity treatment. The following are candidate/control ratios over 80 paired workloads; their scope is a sensitivity model, not measured complete cost.

Trace	Cover	Symmetry	At 0 GAE/native unit	At 0.1	At 1
0	0	0	0.917	83.741	505.962
0	0	1	7.828	32.756	159.835
0	1	0	0.987	19.828	35.753
0	1	1	7.094	9.470	11.479
1	0	0	0.918	61.211	170.195
1	0	1	7.394	23.951	53.878

The trace+cover rows have only eight supported workloads and cannot be ranked as complete 80-workload comparisons. Every arm's recorded unpriced counter has a count. The ordering changes within the pricing band for the apparent lower-bound reductions, so no IC advantage is admitted. The native tool's wall-admission label under a requested L0 minimum is excluded by the stricter experiment protocol; its wall output is diagnostic only.

Vector figures and editable data accompany this report on the following pages.

Formal checks and validation

The proof map pins Lean 4.34.1 and Mathlib. Five exact cleared polynomial identities and four conditional trace/eigenspace statements on Mathlib's actual Weierstrass point group build successfully. The repository auditor checks 13 declarations, including four generated defining equations: zero failures, zero compiler-trusted proofs, standard foundations only.

The concrete finite-field Frobenius/twist embedding, odd subgroup, field/tower/chart/Jacobian maps and cost/complexity bridges remain formally open. The GP characteristic-polynomial check does not independently replay arbitrary-divisor norm code. None of these gaps is replaced with an axiom.

The final root library suite passed **4,479 tests**, with zero failures and 119 existing ignored tests, including the new scalar-Frobenius regression. The standalone suite passed 128 supporting tests; the repeated-eigenvalue amendment additionally executes every-point dual and scalar-return checks. Earlier sparse-fixture/socket failures, the stale 1,324-model roster assertion and invalid CAS/printing attempts are retained. The roster now expects 1,358 extension models, preserving every per-model field/subgroup check. The three backlog cards validate and remain blocked for the full campaign's unmet criteria. Conductor was unavailable locally.

Requirement status

ID	Bounded execution	Remaining original requirement
E00	Five identities and false controls; five Lean polynomial proofs	Group/map bridges are separate
E01	Nine exhaustive trace/fiber controls	General formal Frobenius/twist bridge
E02	Coordinate/slope/norm restrictions executed	Full larger-field norm-y/power families
E03	30 exact rational-quartic witness comparisons	Cost comparison beyond equivalence
E04	Norm controls and projected bases	Complete larger-field fibers and full costing

ID	Bounded execution	Remaining original requirement
E05	Two sparse-function/divisor controls	Unsplit degree-three/four factors and broader divisors
E06	1,179 configured finite-base census rows	Broad timed/count-priced decomposition comparison
E07	240 distinct projection cells, exceptional branches retained	B=12/16 and other base families
E08	Three preserved 1,930-cell bounded attempts	Applicable signature/F5 reference, broader grid and full solver cost
E09	Three translated two-torsion controls per fixture	General proof bridge
E10	Exhaustive declared cover-family search on ten fixtures	No universal cover classification claimed
E11	Toy filters plus five NIST module filters; P-521 passes	Positive module cases still need torsion bridges, quotients and polarization/maps
E12	Ten class-transfer checks, 5,855 classes and independent orders	Faithful formal/CAS arbitrary-class bridge and independent host
E13	Declared low-height image controls	Jacobian decomposition solver and useful-row full cost
E14	Three explicit maps and three verified duals in V2	Complete IC cost/representation comparison
E15	1,920 records; 1,704 verified and exactly replayed	Calibrated complete cost, broader factorial and independent host
E16	Fifteen public-parameter checks; known positive pairing control	No applicable NIST transfer supplied
E17	Integral, standard-height and equal-original-height searches	One rank interval, general target lifting and full discovery cost

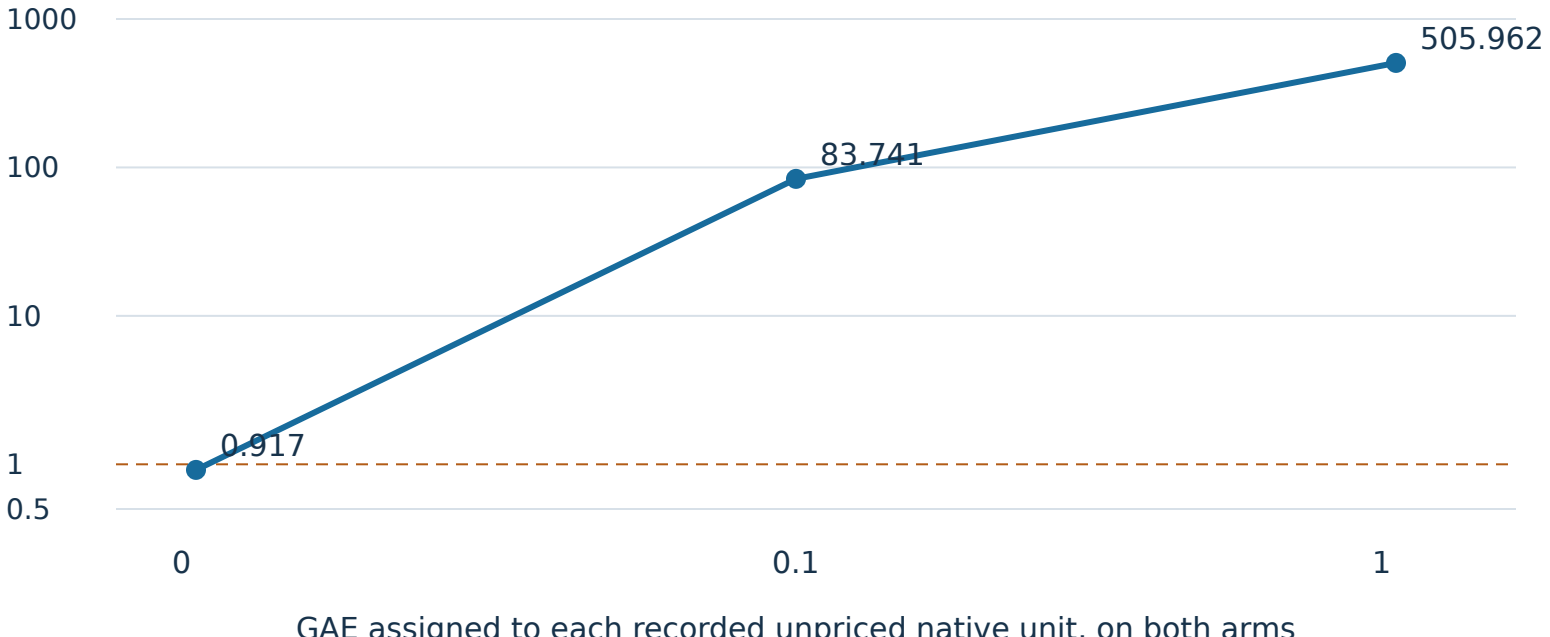
The run reports include each failed attempt, editable quantitative figures and PDFs. Large raw solver trees are losslessly packaged as deterministic split archives; PARI scripts/stdout/stderr have raw archives and member hashes. Original committed evidence is preserved.

Canonical leaderboard, frontier, progress plots and benchmark result rows were checked for impact and left unchanged: this PR supplies no eligible calibrated result for promotion. The curve registry, alias mirror, native catalog views and browser were refreshed. The supersingular pairing-control extension exposed a zero-discriminant trait bug; the fix marks quadratic CM/volcano data inapplicable and has a regression test. No remote lease, funded campaign or follow-on job remains running.

Effect of pricing the recorded native work

t0c0s0 / same-base-t0c0 · 80 paired public workloads · L0

Sensitivity model; no calibrated full-cost ranking or wall-clock result



Y axis: candidate/reference cost ratio (log scale). Dashed line: equal modeled cost.

Verified rational isogeny maps; no cost comparison or assigned volcano depth

